

Data-Driven Modeling and Analysis of a Gas-Steam Combined Cycle

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Introduction

This study builds a thermodynamic analysis framework for a 2-on-1 gas-steam combined heat and power (CHP) unit based on field operation data (April–December 2025). The framework covers working-fluid modeling, gas-turbine and steam/water-side energy analysis, and whole-plant efficiency-factor analysis under both power-only and heating modes.

1. Working-Fluid Modeling

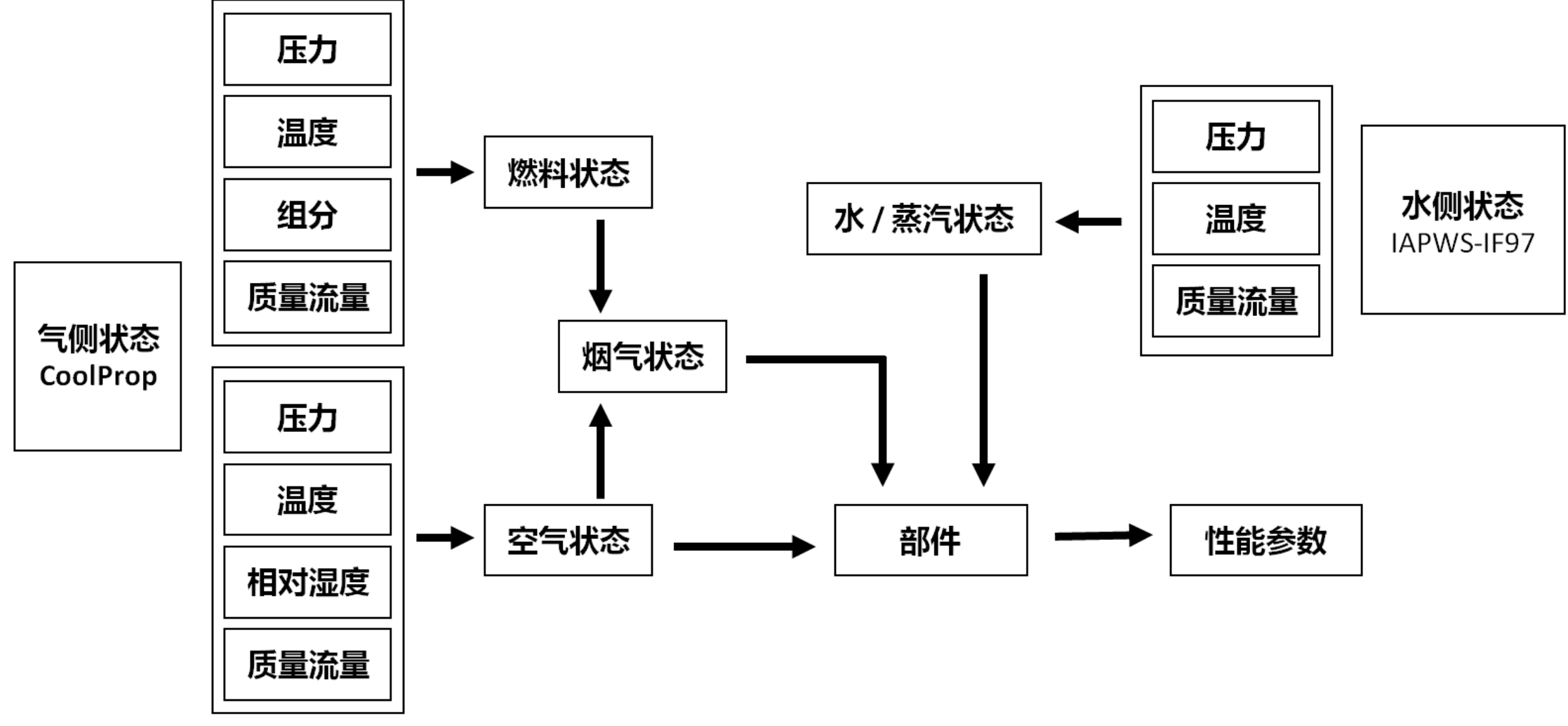


Fig. 1. Working-fluid state and property modeling.

Gas, air, fuel, water, and steam streams are represented by unified flow-state objects. Thermodynamic properties such as enthalpy, entropy, energy flow, and exergy flow are calculated consistently for downstream component models.

2. Gas-Turbine Analysis

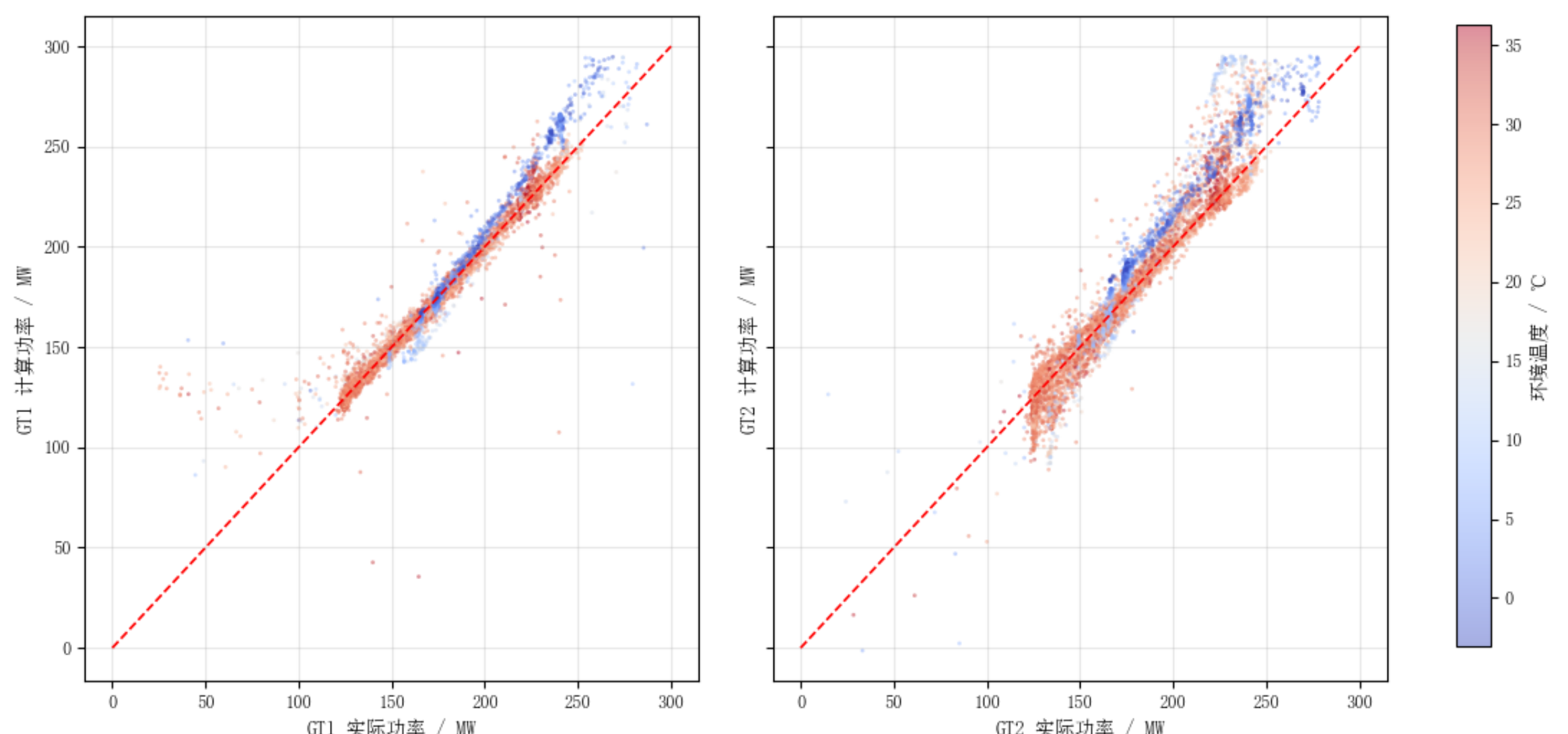


Fig. 2. Gas-turbine calculated and measured power.

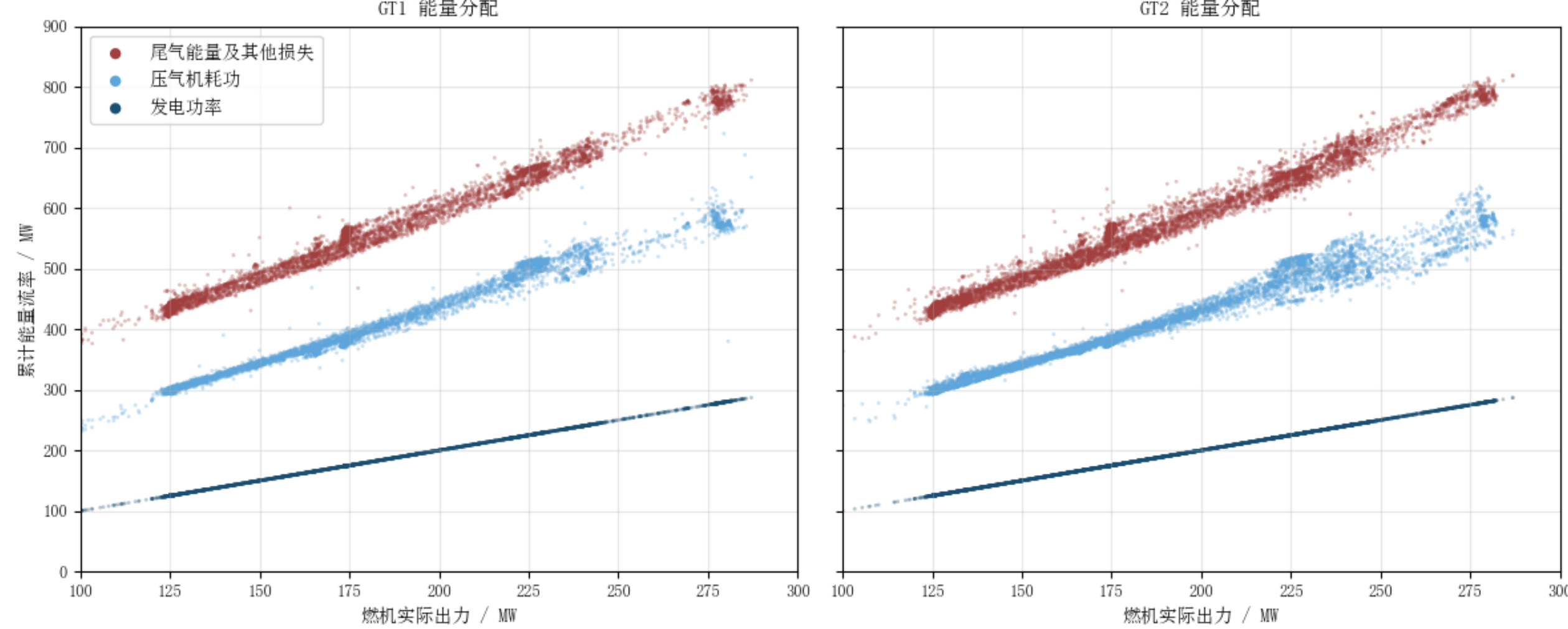


Fig. 3. Fuel-energy distribution in the gas turbine.

The gas-turbine model evaluates compressor work, turbine work, net power, fuel energy, and exhaust heat. The energy distribution clarifies how fuel input is divided into electric output, internal compression work, and recoverable exhaust energy.

3. Steam/Water-Side Analysis

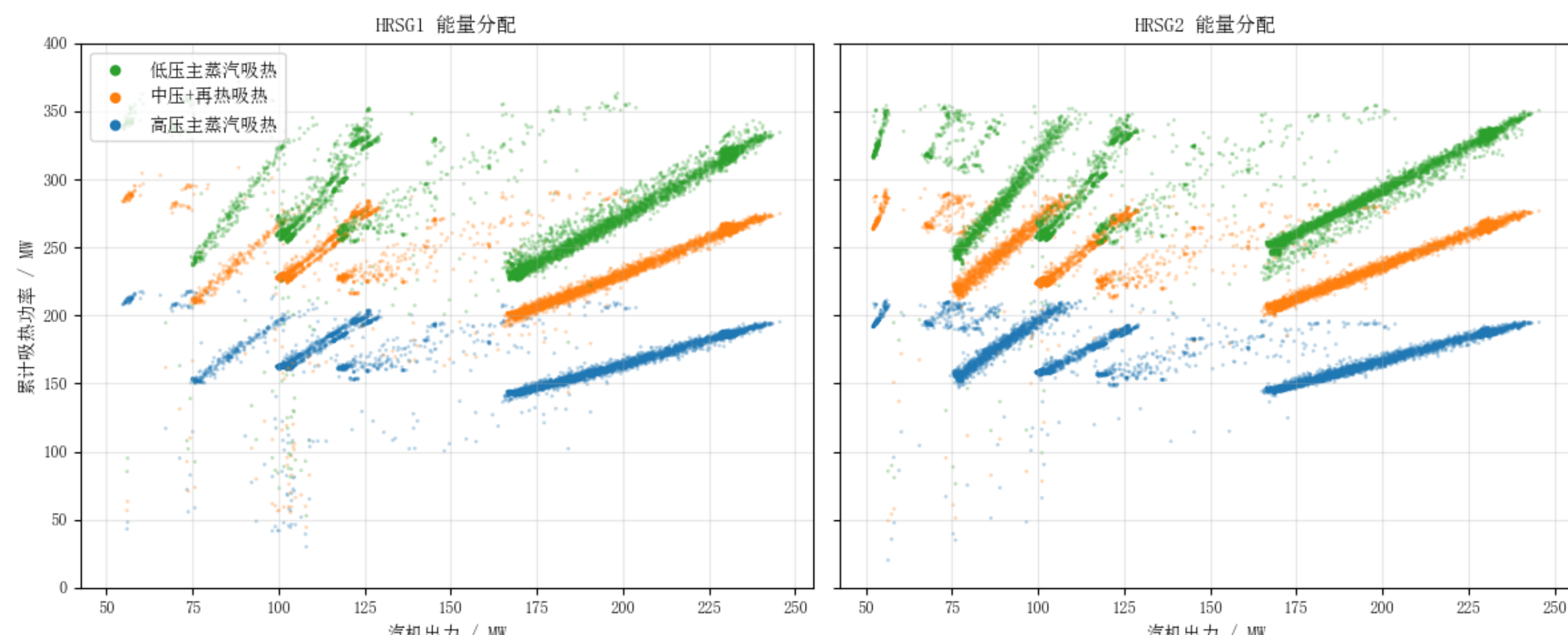


Fig. 4. Heat absorption distribution on the steam/water side.

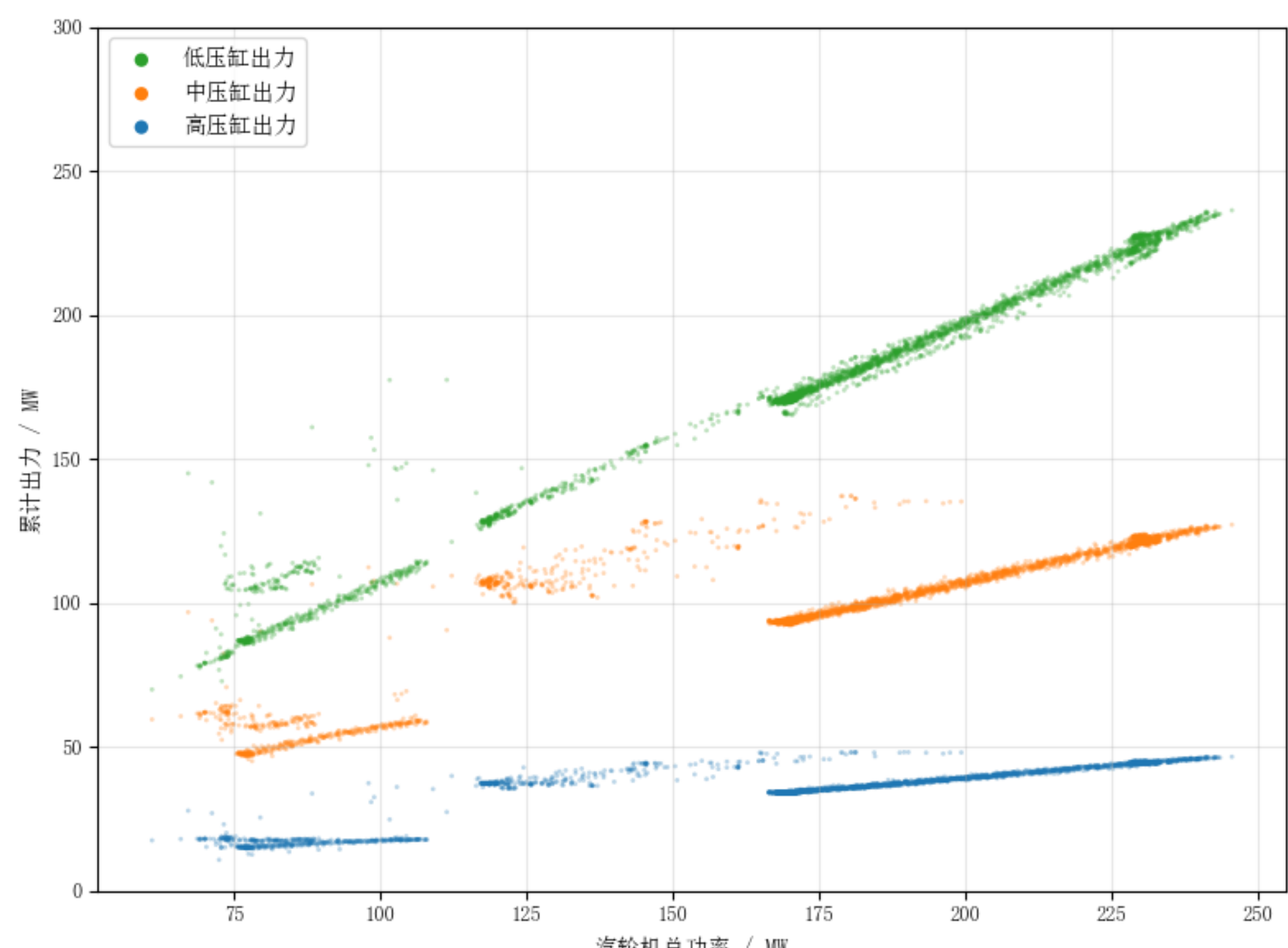


Fig. 5. Power distribution among HP, IP, and LP cylinders.

The HRSG transfers gas-side heat to high-pressure steam, intermediate-pressure/reheat steam, and low-pressure steam. The steam turbine further converts steam energy into cylinder power, while heating extraction changes the effective output definition.

4. Whole-Plant Efficiency

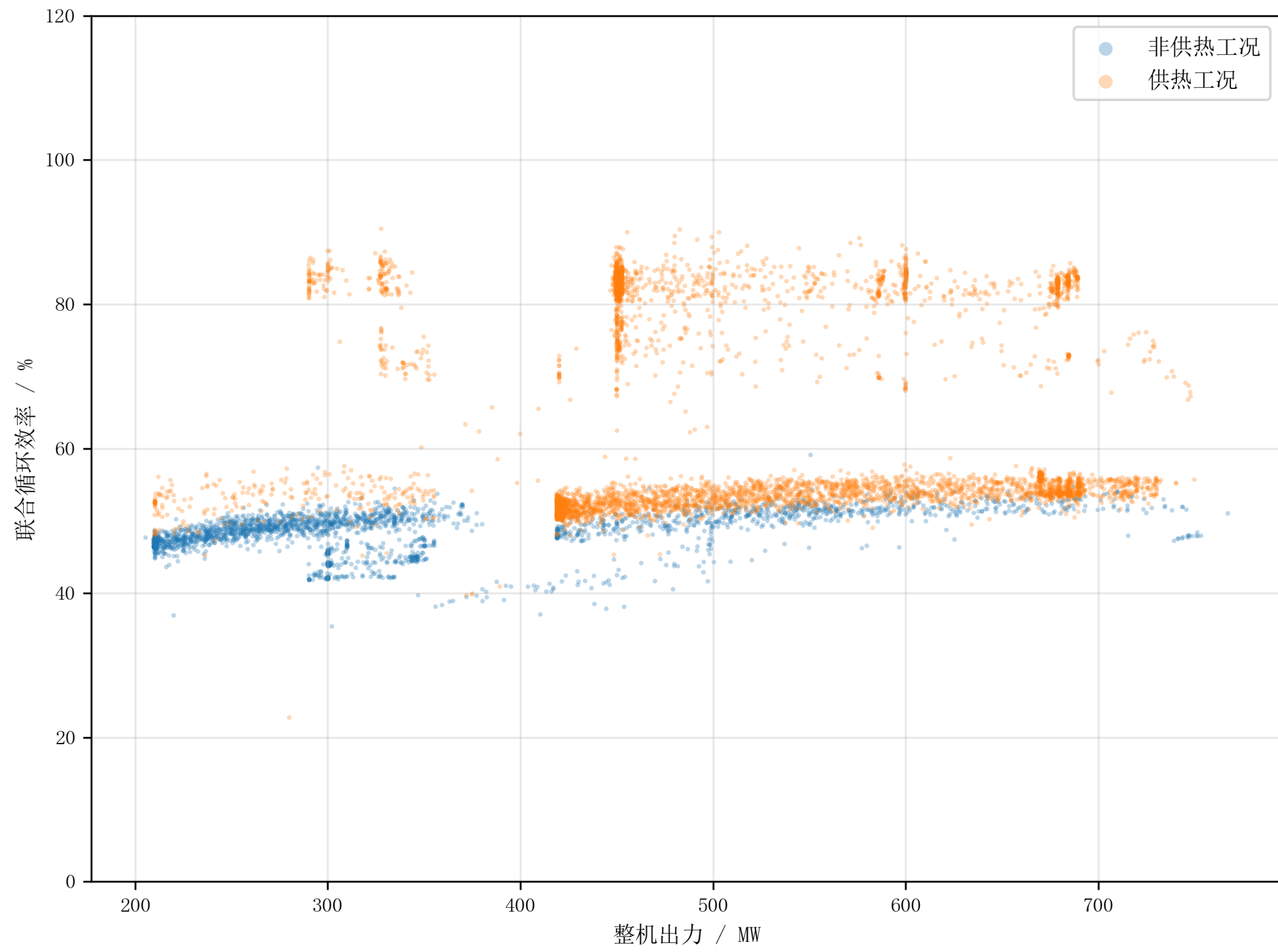


Fig. 6. Combined-cycle efficiency versus total plant power.

Power-only samples cluster by single-GT versus dual-GT operation. Heating samples reach higher utilization because CHP efficiency counts heat supply as useful output: $\eta_{\text{CHP}} = (W_e + Q_h)/Q_F$.

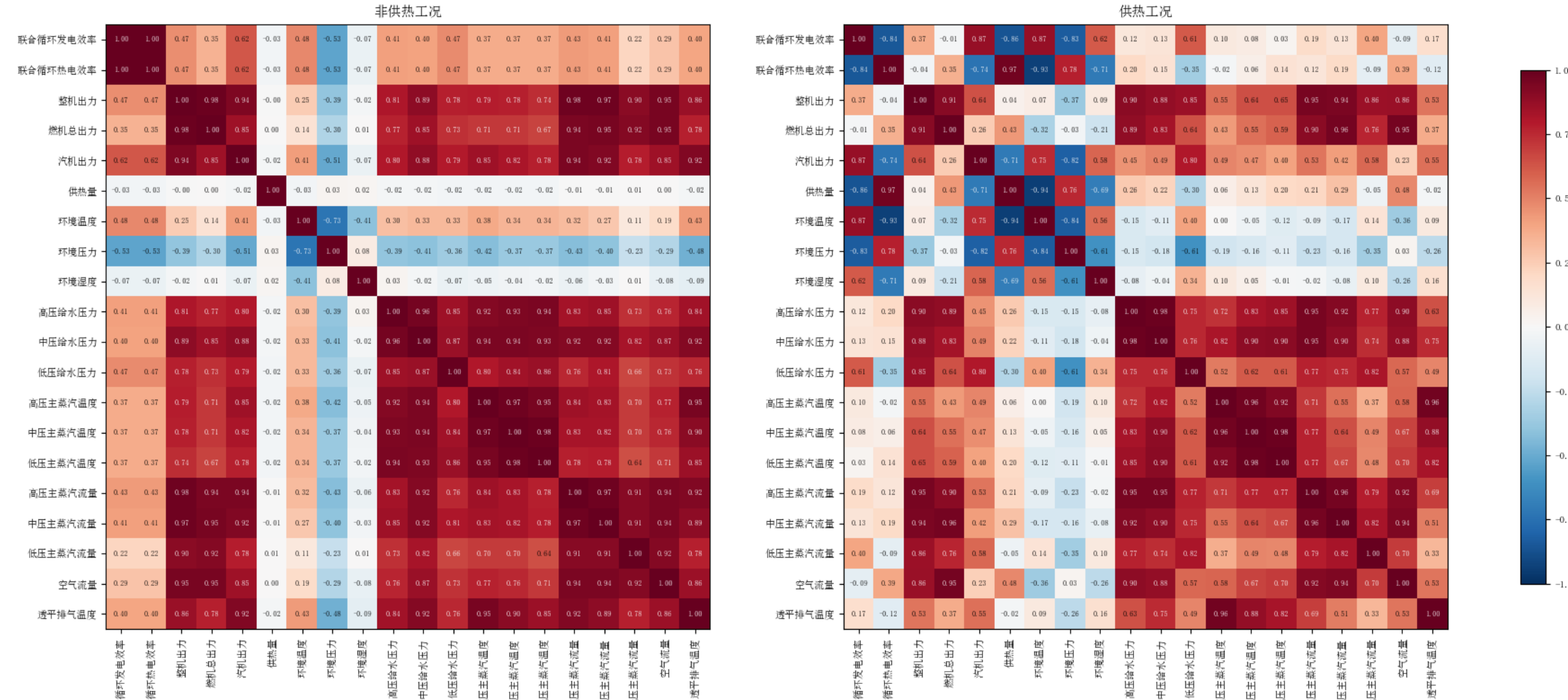


Fig. 7. Pearson correlation matrices for power-only and heating modes.

The two modes are analyzed separately: power-only uses electric efficiency while heating uses CHP efficiency, since heat supply changes the numerator of the efficiency definition and reverses the sign of some correlations.

5. Key Efficiency Factors

Pearson correlation and scatter-plot analyses identify the dominant factors for each mode. In power-only operation, efficiency correlates most with ST output, ambient pressure, and HP main-steam flow. In heating operation, CHP efficiency is dominated by heat supply ($r = 0.966$) and ambient temperature ($r = -0.930$); ST output is negatively correlated ($r = -0.739$) because increased generation reduces heating extraction.

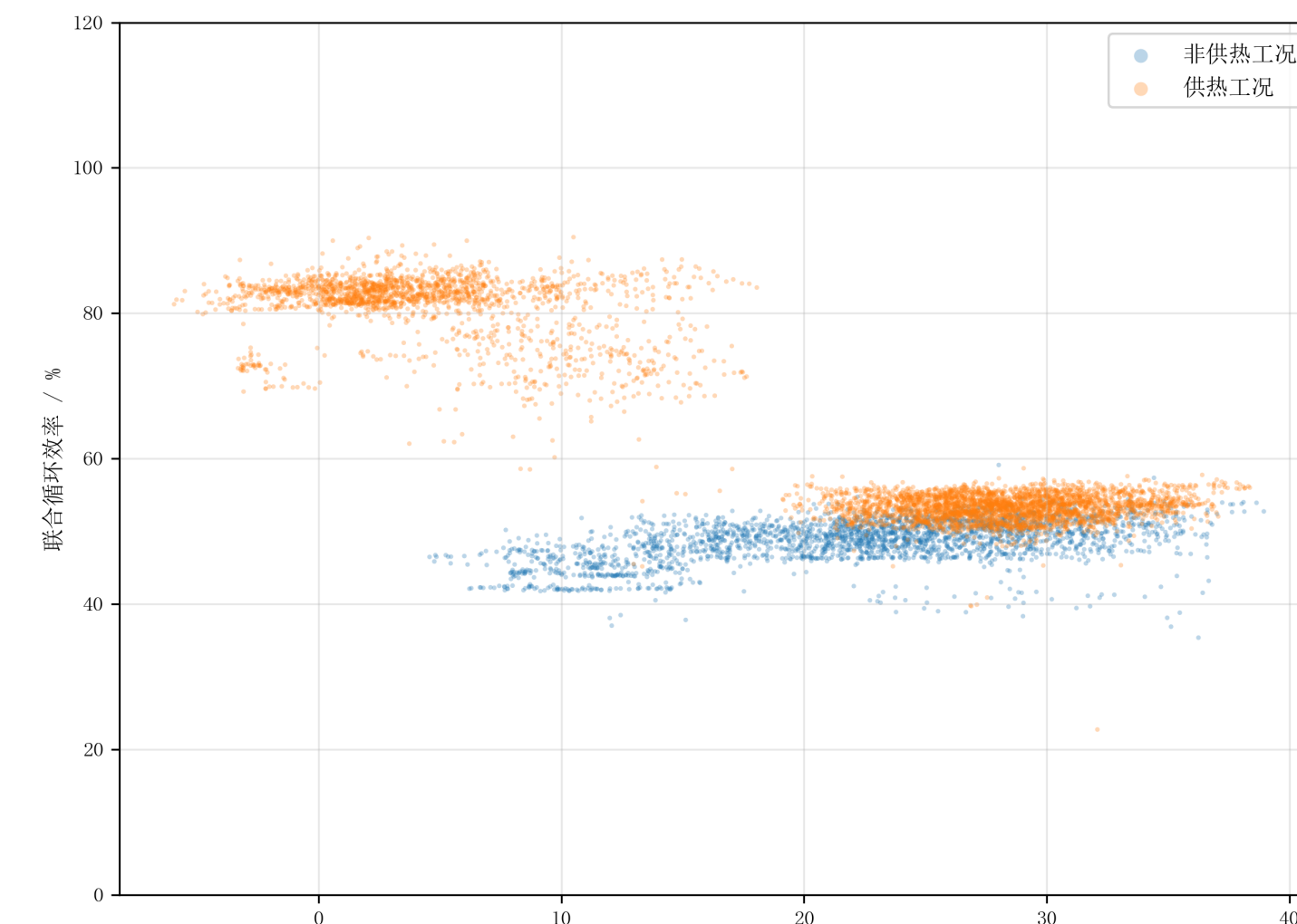


Fig. 8. Ambient temperature vs. efficiency.

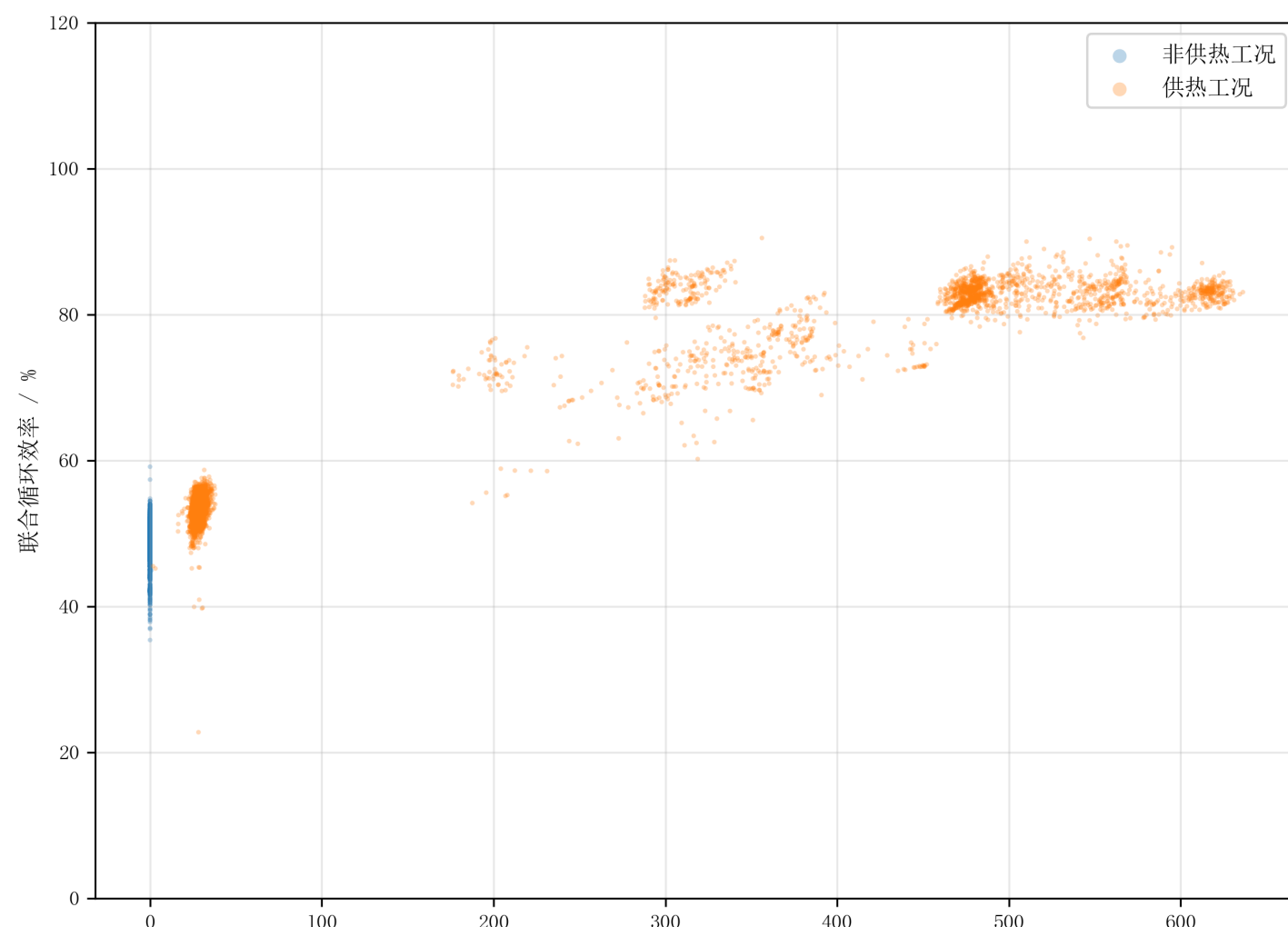


Fig. 9. Heat supply vs. efficiency.

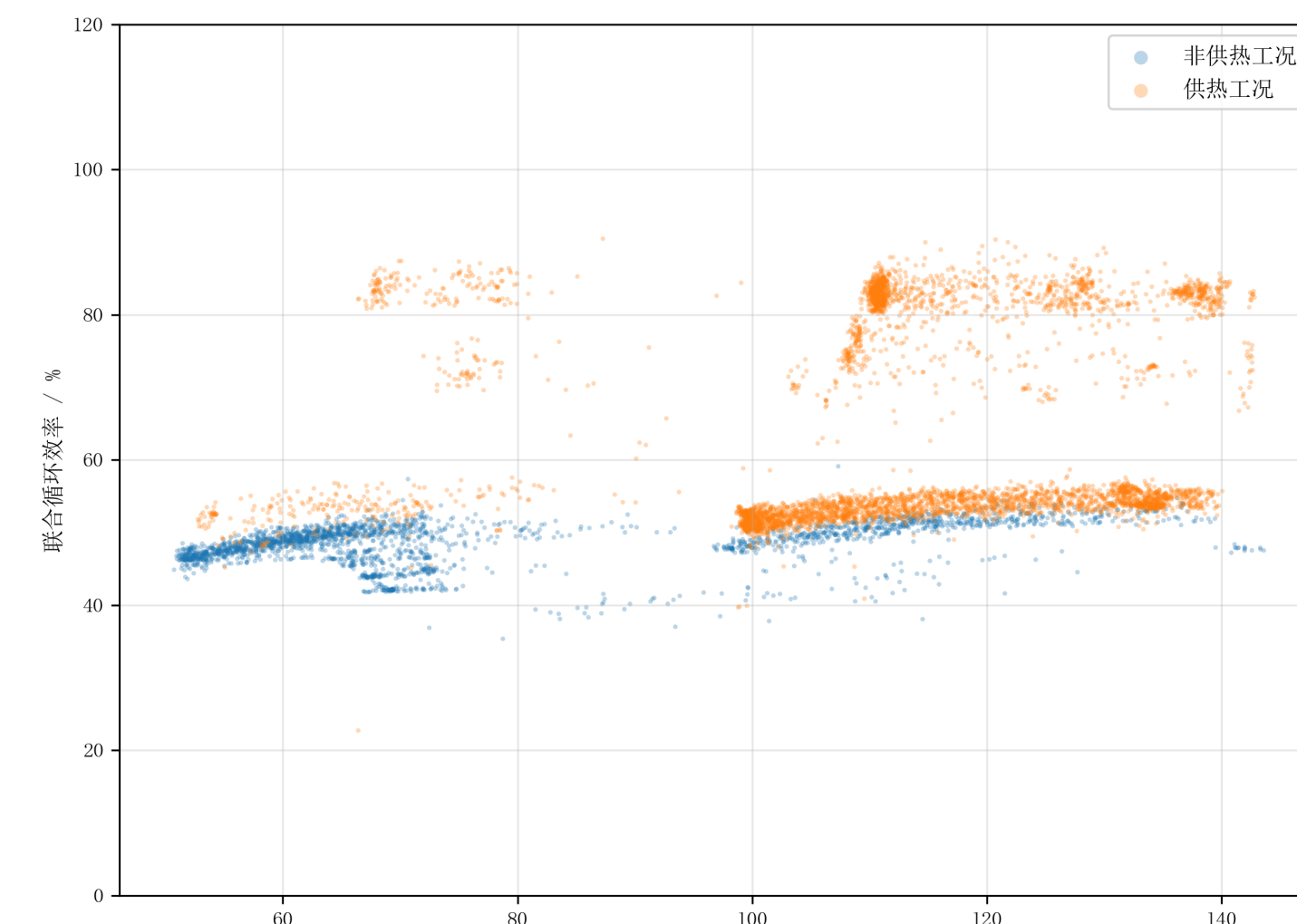


Fig. 10. HP main-steam flow vs. efficiency.

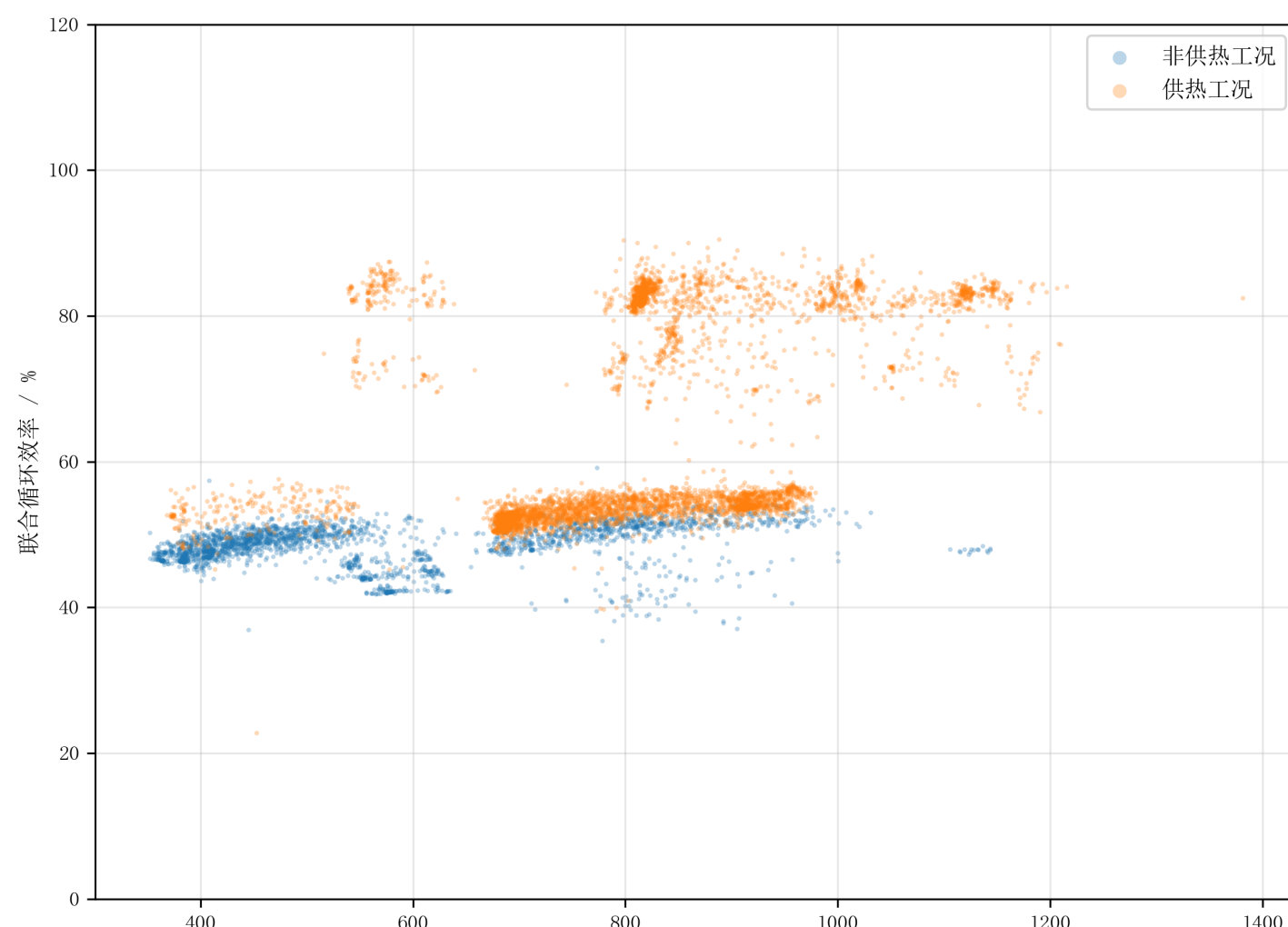


Fig. 11. Air flow vs. efficiency.

6. Summary and Future Work

- A unified flow-state framework supports component-level and plant-level energy analysis from field data.
- GT analysis quantifies fuel-energy distribution among power output, compressor work, and exhaust heat.
- HRSG and ST analysis links water-side heat absorption with cylinder power and heating extraction.
- Power-only and heating modes require separate efficiency definitions;